MedScribe AI

Mobile-First Ambient AI Scribe
with Real-Time FHIR Conversion

Problem Statement: PS-1

Hackathon: HACKMATRIX 2.0

Organized by: Jilo Health × NJACK, IIT Patna

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PROPOSED SOLUTION

The Problem

India's healthcare system serves over 1.4 billion people, yet its clinical workforce remains critically understaffed. The doctor-to-patient ratio stands at roughly 1:1,000, well below the WHO recommendation of 1:1,000 in urban areas and far worse in rural settings. In this high-pressure environment, physicians spend an estimated **2–3 hours every day** on clinical documentation—writing case notes, filling discharge summaries, coding diagnoses, and preparing referral letters. This administrative burden directly reduces the time available for patient care, accelerates burnout, and contributes to India's growing physician attrition crisis.

Compounding the problem is India's unique linguistic landscape. Doctor-patient conversations routinely occur in **Hindi-English code-mixed speech**—a patient might describe their symptoms as *"pet mein dard ho raha hai, 3 din se fever bhi hai"* while the doctor responds with clinical English interspersed with Hindi reassurances. Global ambient scribe tools (Abridge, Nuance DAX, Suki) are built for monolingual English clinical settings and **fail catastrophically** on code-mixed Indian conversations. Meanwhile, India's ambitious **Ayushman Bharat Digital Mission (ABDM)** mandates FHIR-compliant health records for interoperability, yet no affordable tool exists that can convert a spoken consultation into standards-compliant digital records. Rural telemedicine expansion—a national priority—remains bottlenecked by the documentation overhead imposed on already-overstretched physicians.

Our Solution

MedScribe AI is a mobile-first Progressive Web App that transforms spoken doctor-patient conversations into structured clinical notes and FHIR R4–compliant resources in real time. The workflow is radically simple: a doctor opens the app on their phone, taps a single record button, and speaks naturally with their patient in any mix of Hindi and English. As the conversation flows, MedScribe AI streams a live transcript while simultaneously extracting chief complaints, diagnoses with ICD-10 codes, medications with dosage and RxNorm mappings, vitals coded to LOINC, allergies, and a structured SOAP note—all updated incrementally on screen.

What sets MedScribe AI apart is the integration of **Clinical Decision Support** directly into the documentation pipeline. As medications and allergies are extracted, the system checks for drug-drug interactions, allergy cross-reactivity, and dosage safety—raising real-time alerts before the prescription is finalized. A **Differential Diagnosis Engine** suggests alternative diagnoses with supporting evidence and distinguishing tests. Every consultation produces a scored FHIR R4 Bundle (Grade A–D) ready for ABDM integration, complete with Patient, Encounter, Condition, Observation, MedicationRequest, AllergyIntolerance, CarePlan, and ServiceRequest resources. The system learns continuously: when a doctor corrects an extraction, that correction is stored and injected as a few-shot example into future calls, improving accuracy without model retraining.

MedScribe AI goes beyond English and Hindi. With **Sarvam AI integration** for server-side speech-to-text, the system now supports **10 Indian languages** including Tamil, Telugu, Bengali, Marathi, Gujarati, Kannada, and Malayalam—with full code-mixed support (Hinglish, Tanglish, and more). Doctors can toggle between the zero-cost browser STT and Sarvam AI's superior accuracy for Indian languages. A **digital prescription system with QR codes** lets patients scan and view their prescription on any phone, while a **Patient Safety Score** (0–100) provides an at-a-glance clinical risk assessment. A **patient consent recording** mechanism ensures GDPR-style compliance before any recording begins, and **13 clinical nudge rules** prompt doctors when critical history elements are missing from the conversation.

System Architecture

Data Flow

The browser captures speech via either the native Web Speech API or **Sarvam AI's server-side STT** (supporting 22 Indian languages with code-mixed speech). Doctors select their preferred language and STT engine from an in-app selector. Recognized text streams over a WebSocket to the FastAPI backend. At session end, the full transcript is sent to Gemini 2.5 Flash in a single **batch extraction** call—eliminating data loss from incremental overwrites. The AI returns structured JSON which is mapped to FHIR R4 resources, scored for completeness, and run through the CDS engine for safety checks. All results stream back to the frontend over the same WebSocket connection.

Technology Stack

Layer	Technology	Rationale
Frontend	React 18, Vite, Tailwind CSS	PWA-capable, fast HMR, mobile-first utility CSS
Speech-to-Text	Web Speech API + Sarvam AI	Dual-mode: free browser STT or Sarvam AI for 22 Indian languages
Backend	Python FastAPI + WebSocket	Async-native, auto-docs, WebSocket first-class
AI Extraction	Google Gemini 2.5 Flash	Fast structured output, code-mixed understanding
FHIR Mapping	Custom Python + Pydantic models	Type-safe R4 resource generation with validation
CDS Engine	Rule-based Python service	Deterministic safety checks, no AI hallucination risk
Terminology	ICD-10, SNOMED CT, RxNorm, LOINC	International coding standards, ABDM alignment

Security & Privacy Considerations

- Speech processed in-browser (Web Speech API) or via encrypted API call (Sarvam AI)—raw audio never stored
- **GDPR-style patient consent recording** required before any recording session begins
- WebSocket connection scoped to session; no persistent patient data stored on server
- FHIR resources generated on-demand, exportable as local JSON files
- **Doctor registration & authentication** with hospital affiliation and session persistence
- No patient identifiers stored in correction/learning data
- PWA supports offline-capable UI shell via service worker caching

Key Features

FEATURE 01

10 Indian Languages + Code-Mixed Speech

Supports Hindi, English, Tamil, Telugu, Bengali, Marathi, Gujarati, Kannada, and Malayalam with full code-mixed support (Hinglish, Tanglish). Dual STT: browser-native Web Speech API (free) or **Sarvam AI** server-side STT with 22 Indian languages. Maps **59 Indian drug brand names** to generic equivalents with RxNorm codes and standard dosages.

FEATURE 02

Clinical Decision Support (CDS)

Real-time safety engine with **10 drug-drug interaction rules**, **3 allergy cross-reactivity checks**, and **10 dosage validation rules**. Catches dangerous combinations (e.g., Warfarin + Aspirin, NSAID in renal impairment) before the prescription is finalized. Severity-graded alerts (high / moderate / low).

FEATURE 03

FHIR R4 + ABDM Compliance

Generates a complete FHIR R4 Bundle with **8 resource types**: Patient, Encounter, Condition (ICD-10), Observation (LOINC), MedicationRequest (RxNorm), AllergyIntolerance, CarePlan, and ServiceRequest. Each resource is quality-scored (Grade A–D) for coding completeness and inter-resource referential integrity.

FEATURE 04

Differential Diagnosis Engine

For every primary diagnosis, the AI suggests **2–3 alternative diagnoses** with supporting evidence extracted from the conversation and recommended distinguishing tests. Helps doctors consider conditions they might overlook, supporting diagnostic rigor and reducing misdiagnosis rates.

FEATURE 05

Continuous Learning from Corrections

Doctors can edit any extracted clinical note field and click **"Save & Teach AI."** Corrections are stored and injected as few-shot examples into future Gemini extraction calls. The system improves with use—no model fine-tuning or retraining required. Accuracy compounds across consultations.

FEATURE 06

Digital Prescription with QR Code

End-of-session QR code generation linking to `/rx/{sessionId}`. Patient scans the code to view a formatted prescription page with patient info, medications table, diagnosis, and follow-up instructions—accessible on any smartphone without app installation.

FEATURE 07

Patient Safety Score

A composite **0–100 clinical risk score** computed from CDS alerts, missing history elements, allergy checks, and medication safety. Displayed as an animated SVG visualization with color-coded severity (green/amber/red), giving doctors an instant at-a-glance risk assessment.

FEATURE 08

Clinical Nudges & Batch Extraction

13 AI-powered nudge rules prompt the doctor for missing history elements (e.g., allergy history, family history, medication reconciliation). **Batch extraction** sends the full transcript to Gemini in one shot at session end, eliminating data loss from incremental overwrites.

Additional Capabilities

Capability	Description
Doctor Registration	Login screen with doctor name, ID, hospital affiliation. Session persistence across visits
Patient Consent	GDPR-style consent banner before recording starts; ensures regulatory compliance
59 Drug Brand Mappings	Comprehensive Indian drug reference with brand-to-generic mapping, standard dosages, and use cases
FHIR Quality Scoring	Automated Grade A–D scoring checking resource completeness, coding systems, and cross-references
Export Suite	Print-ready notes, FHIR Bundle JSON download, clipboard copy, and QR-based digital prescriptions
Dark Mode + PWA	Full dark theme for low-light clinical environments; installable on mobile home screen

Competitive Differentiation

Capability	MedScribe AI	Abridge	Nuance DAX	Suki AI
Indian Languages (10+)	✔ 10 languages	✘ English only	✘ English only	✘ English only
ABDM / FHIR R4 Alignment	✔ Full R4 Bundle	● Partial	● Via Epic	● Partial
Clinical Decision Support	✔ Built-in CDS	✘ None	✘ External	✘ None
Indian Drug Brand Mapping	✔ 59 brands mapped	✘	✘	✘
Differential Diagnosis	✔ With evidence	✘	✘	● Basic
Digital Prescription (QR)	✔ QR scannable	✘	● Via EHR	✘
Patient Safety Score	✔ 0–100 composite	✘	✘	✘
Speech API Cost	✔ $0 or free credits	Proprietary	Proprietary	Proprietary
Continuous Learning	✔ Few-shot injection	✘	✘	✘
Pricing (India)	Open source / Free	~$1,000/mo	~$1,500/mo	~$400/mo

Impact Metrics

10
INDIAN LANGUAGES

59
DRUG BRAND MAPPINGS

36
SAFETY RULES (CDS + NUDGES)

8
FHIR R4 RESOURCE TYPES

Projected Clinical Impact

- **Time savings:** Reduces documentation time from ~15 minutes to ~1 minute per consultation, freeing 2+ hours/day for direct patient care.
- **Patient safety:** Real-time CDS catches dangerous drug interactions and allergy cross-reactivity before prescriptions are finalized—errors that manual review often misses under time pressure.
- **Digital health readiness:** Automatic FHIR R4 output means every consultation is ABDM-ready from day one, accelerating India's health record interoperability goals.
- **Equity:** Browser-native speech recognition with zero API cost makes ambient scribing accessible to small clinics and rural telemedicine setups, not just well-funded hospital chains.
- **Diagnostic quality:** Differential diagnosis suggestions and risk factor identification support clinical reasoning, especially for junior doctors and GPs handling unfamiliar presentations.

Live Demo

Available Demo Scenarios

#	Scenario	Highlights
1	Viral Fever (Hindi-English)	Code-mixed transcript, Crocin → Paracetamol mapping, ICD-10 coding
2	Diabetes Follow-Up	Medication review, HbA1c observation, CarePlan with diet instructions
3	Cardiac + Safety Alerts	Warfarin + Aspirin interaction, allergy cross-reactivity, high-severity CDS
4	Live Microphone	Real-time recording with actual doctor-patient conversation

How to Run

```
# Clone and install
$ git clone https://github.com/your-org/medscribe-ai.git
$ cd medscribe-ai
$ npm install

# Set Gemini API key
$ echo "GEMINI_API_KEY=your_key" > backend/.env

# Start both frontend and backend
$ npm run dev

# Open http://localhost:5173 on your phone or browser
```

Future Roadmap

PHASE 1 — DELIVERED (HACKATHON)

Full-Featured MVP

10 Indian languages via Sarvam AI + browser STT, FHIR R4 output, CDS engine, differential diagnosis, digital prescriptions with QR, patient safety score, clinical nudges, doctor registration, batch extraction, demo mode, PWA.

PHASE 2 — Q2 2026

EHR Integration & ABDM Bridge

Direct integration with Indian EHR systems (Bahmni, OpenMRS). ABDM Health Information Provider (HIP) module for national health record exchange.

PHASE 3 — Q3 2026

Specialty-Specific Models

Tailored extraction prompts and drug databases for cardiology, pediatrics, obstetrics, and ophthalmology. Specialty-aware CDS rules and FHIR profiles.

PHASE 4 — Q4 2026

Edge Deployment & Offline Mode

On-device AI inference for areas with unreliable internet. Local Whisper model for offline speech recognition. Full offline-first architecture for rural deployment.

Acknowledgments

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